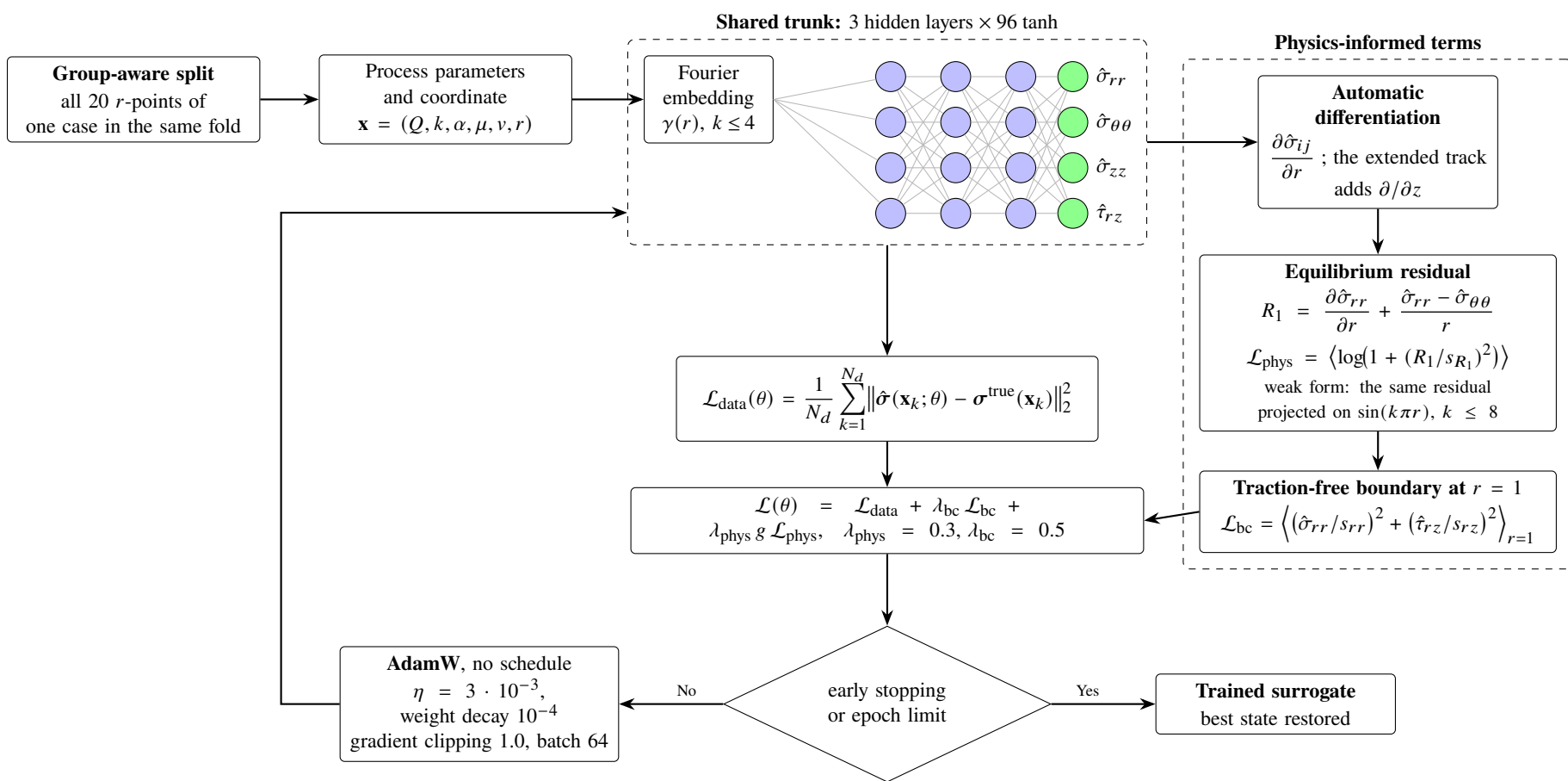




Held fixed for every family and every split. The architecture, the optimiser settings and the loss weights were set before any held-out region was evaluated and were not tuned per family or per split, so no search can leak a held-out region. The only quantity read from the data is the gradient-norm calibration $g = \|\nabla_{\theta} \mathcal{L}_{\text{data}}\|_2 / \|\nabla_{\theta} \mathcal{L}_{\text{phys}}\|_2$, set once on the first mini-batch so that strong and weak forms enter with the same share of the data gradient ($g \approx 0.165$ strong, ≈ 4.97 weak).